KHULNA
UNIVERSITY OF
ENGINEERING AND
TECHNOLOGY

Akash Biswas
Lecturer, EEE
KUET

Department of Electrical and Electronic Engineering

Course Code : EE 4121

VLSI

MASK Layout

Mask Layout

Representation of IC or stick diagram in terms of planar geometric shapes or masks or set of masks is called mask layout.

- Exact geometric information is known
- Chips or IC are specified with mask sets.
- Area is known.

Properties of Mask Layout:

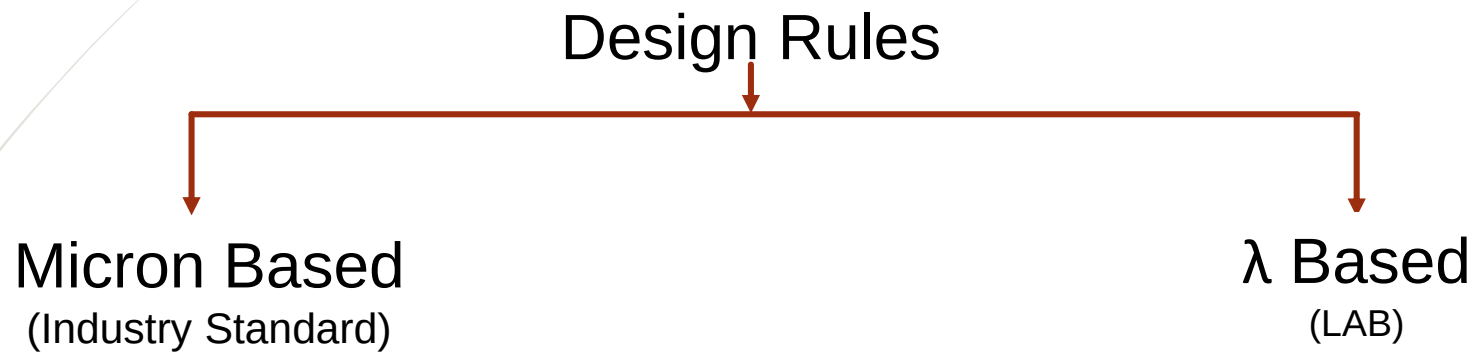
- To ensure minimum size/width –to avoid opening or breaking
- To ensure minimum spacing –to avoid shorting
- To ensure minimum overlapping –for proper functioning
- To ensure minimum surrounding –contact cuts

Design Rules: Design rules are a set of geometrical specifications that dictate the design of the layout

- **Designer :**
  - minimize complexity
  - Utilize area
  - Workable chip
  - Reliable (proper functioning)

- **Process Engineer:**
  - Controllable
  - reproducible

Layout: An interconnection between the designer and the process engineer.



- Industry Standard: Micron Rule
  - All device dimensions are expressed in terms of absolute dimension(μm/nm)
  - These rules will not support proportional scaling

λ Based Design Rules :

- Developed by Mead and Conway.
- All device dimensions are expresses in terms of a scalable parameter λ.

- λ = L/2; L = The minimum feature size of transistor
- L = 2 λ

Distance between Drain and Source

- These rules support proportional scaling.
- They should be applied carefully in sub-micron CMOS process.

- ❖ λ is abstract unit
- ❖ Easy and generalized
- ❖ Widely acceptable

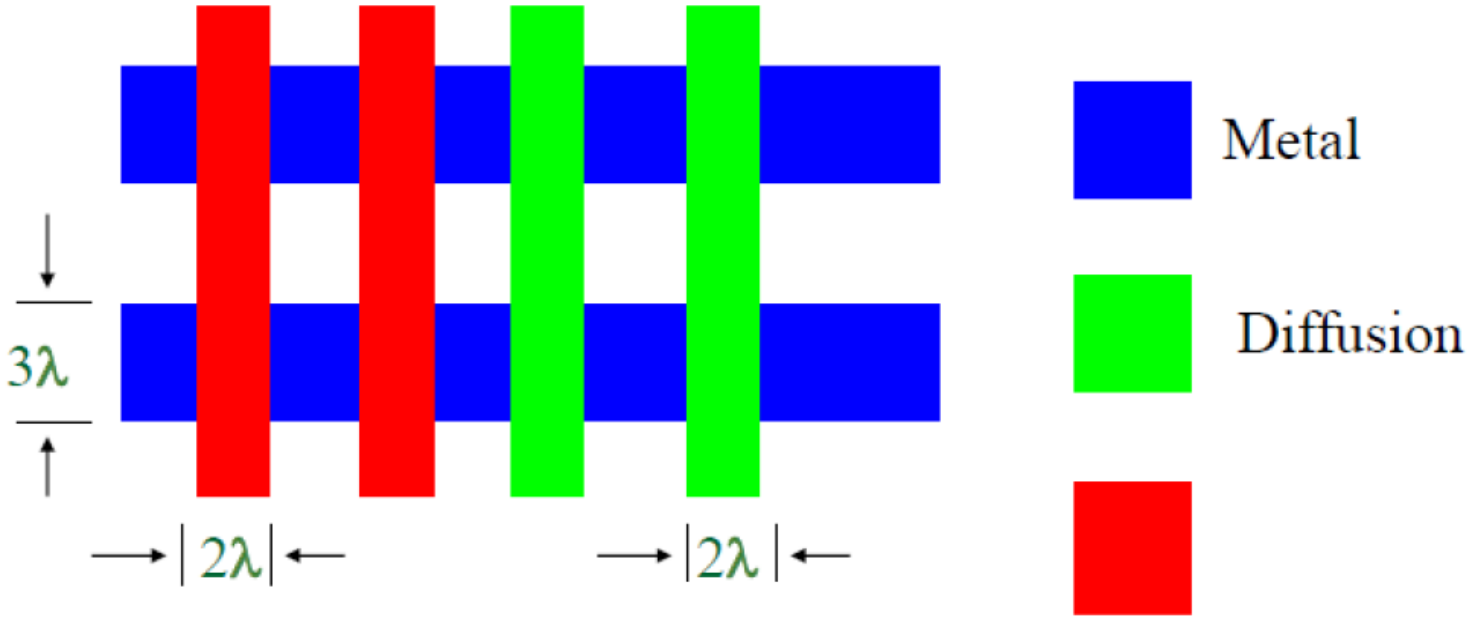
Colour Codes :

Layer	Color	Representation
N+ Active	Green	
P+ Active	Yellow	
PolySi	Red	
Metal 1	Blue	
Metal 2	Magenta	
Contact	Black	X
Buried contact	Brown	X
Via	Black	X
Implant	Dotted yellow	
N-Well	Dotted Green/Black	

Darshana Sankhe

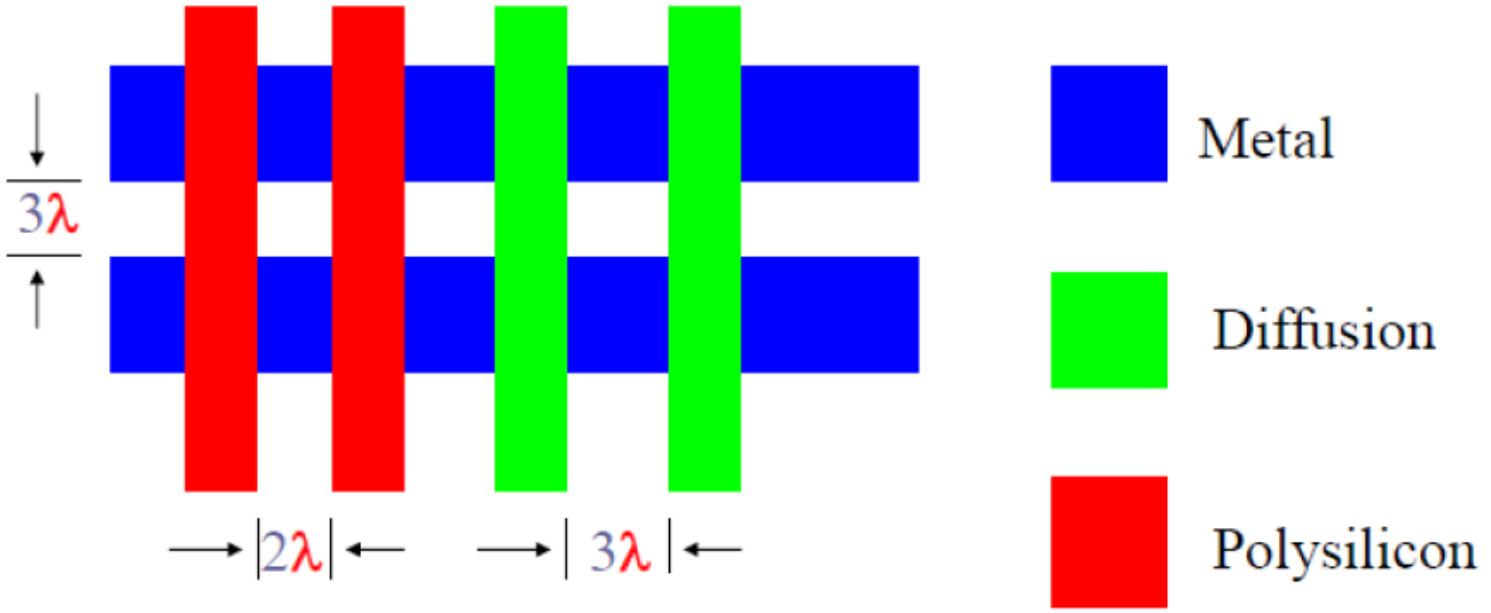
Design Rules:

- Minimum width of PolySi and diffusion line **2λ**
- Minimum width of Metal line **3λ** as metal lines run over a more uneven surface than other conducting layers to ensure their continuity



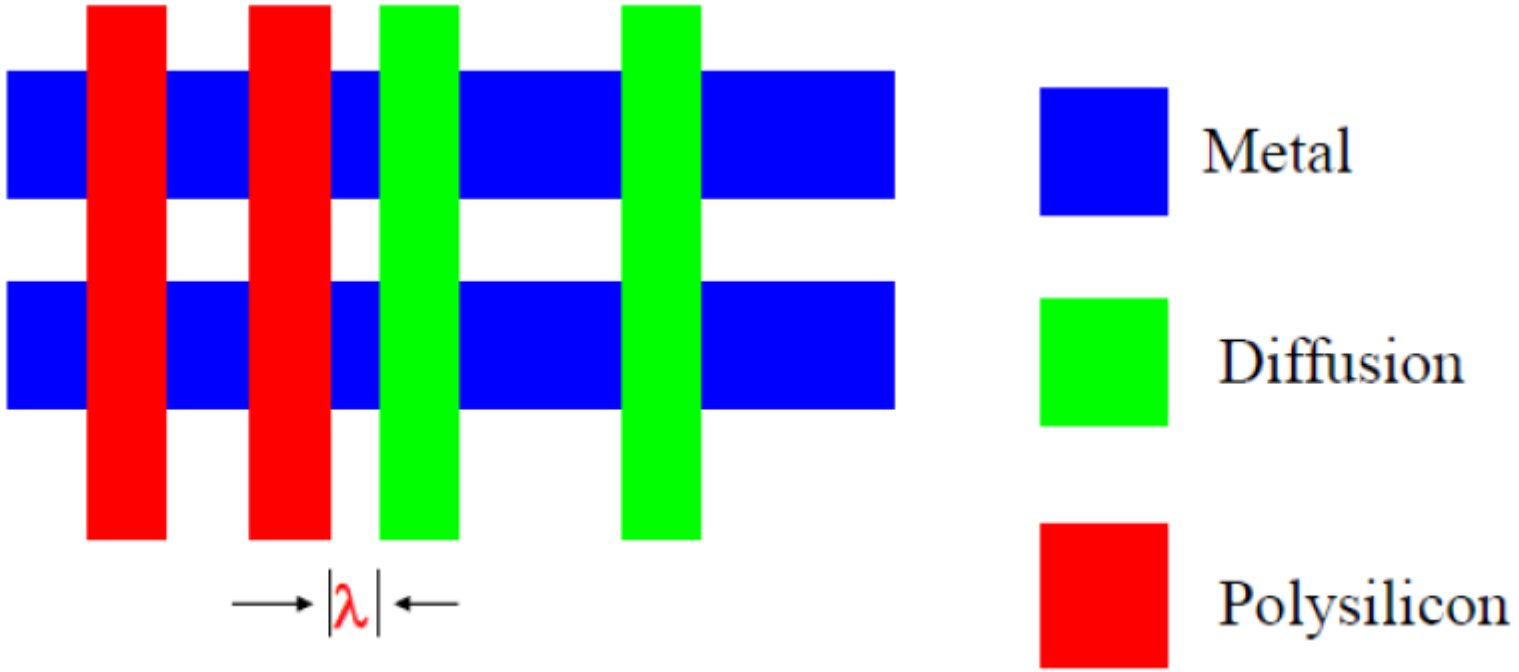
Design Rules:

- PolySi – PolySi spacing 2λ
- Metal - Metal spacing 3λ
- Diffusion – Diffusion spacing 3λ To avoid the possibility of their associated regions overlapping and conducting current



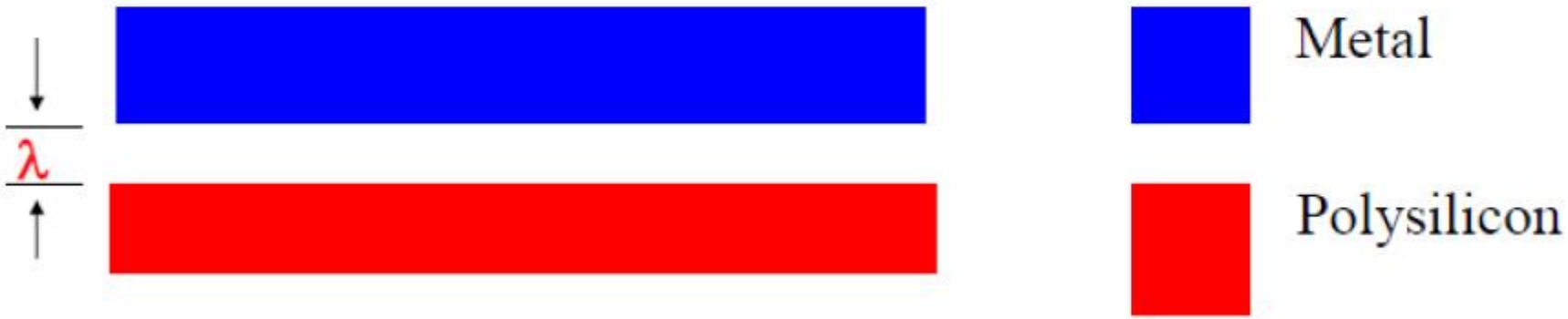
Design Rules:

- **Diffusion – PolySi spacing λ To prevent the lines overlapping to form unwanted capacitor.**
- **Metal lines can pass over both diffusion and polySi without electrical effect. Where no separation is specified, metal lines can overlap or cross.**



Design Rules:

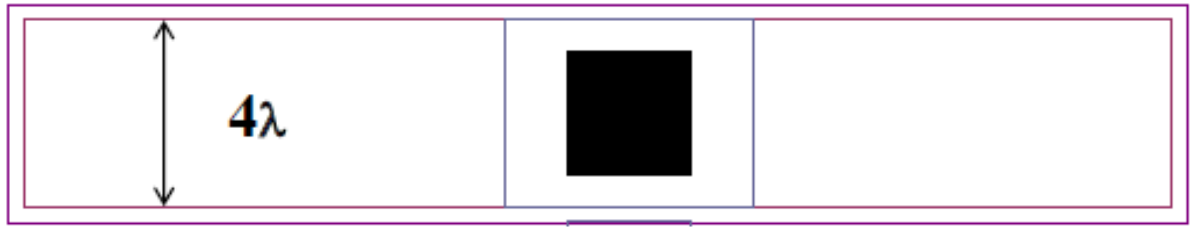
- **Metal lines can pass over both diffusion and polySi without electrical effect**
- **It is recommended practice to leave λ between a metal edge and a polySi or diffusion line to which it is not electrically connected**



Contact Cut :

- Metal connects to polySi/diffusion by contact cut.
- Contact area: 2λ*2λ

- Metal and polySi or diffusion must overlap this contact area by λ so that the two desired conductors encompass the contact area despite any mis-alignment between conducting layers and the contact hole



Design Rules: NMOS

- Minimum diff width 2λ
- Minimum poly width 2λ
- Minimum metal width 3λ
- poly-poly spacing 2λ

- diff-diff spacing 3λ

  (depletion regions tend to spread outward)
- metal-metal spacing 3λ

- diff-poly spacing λ

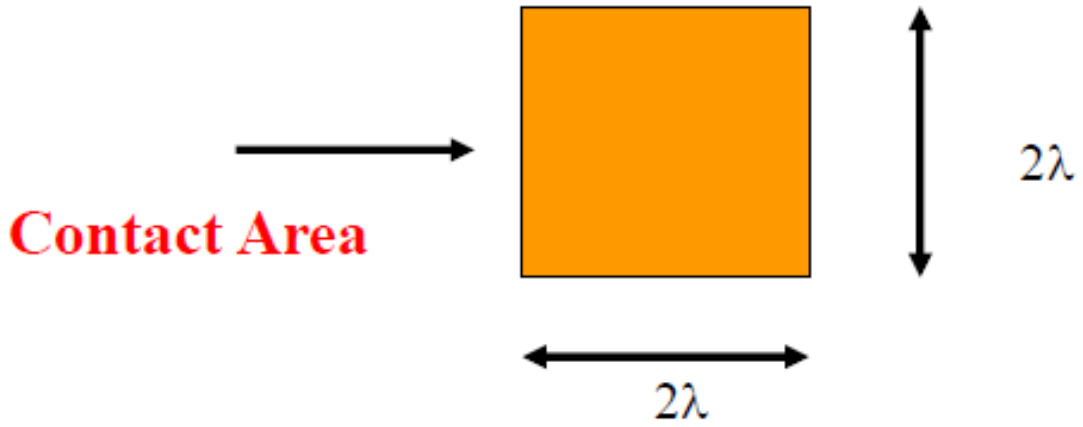
Design Rules: NMOS

- Poly gate extend beyond diff by 2λ

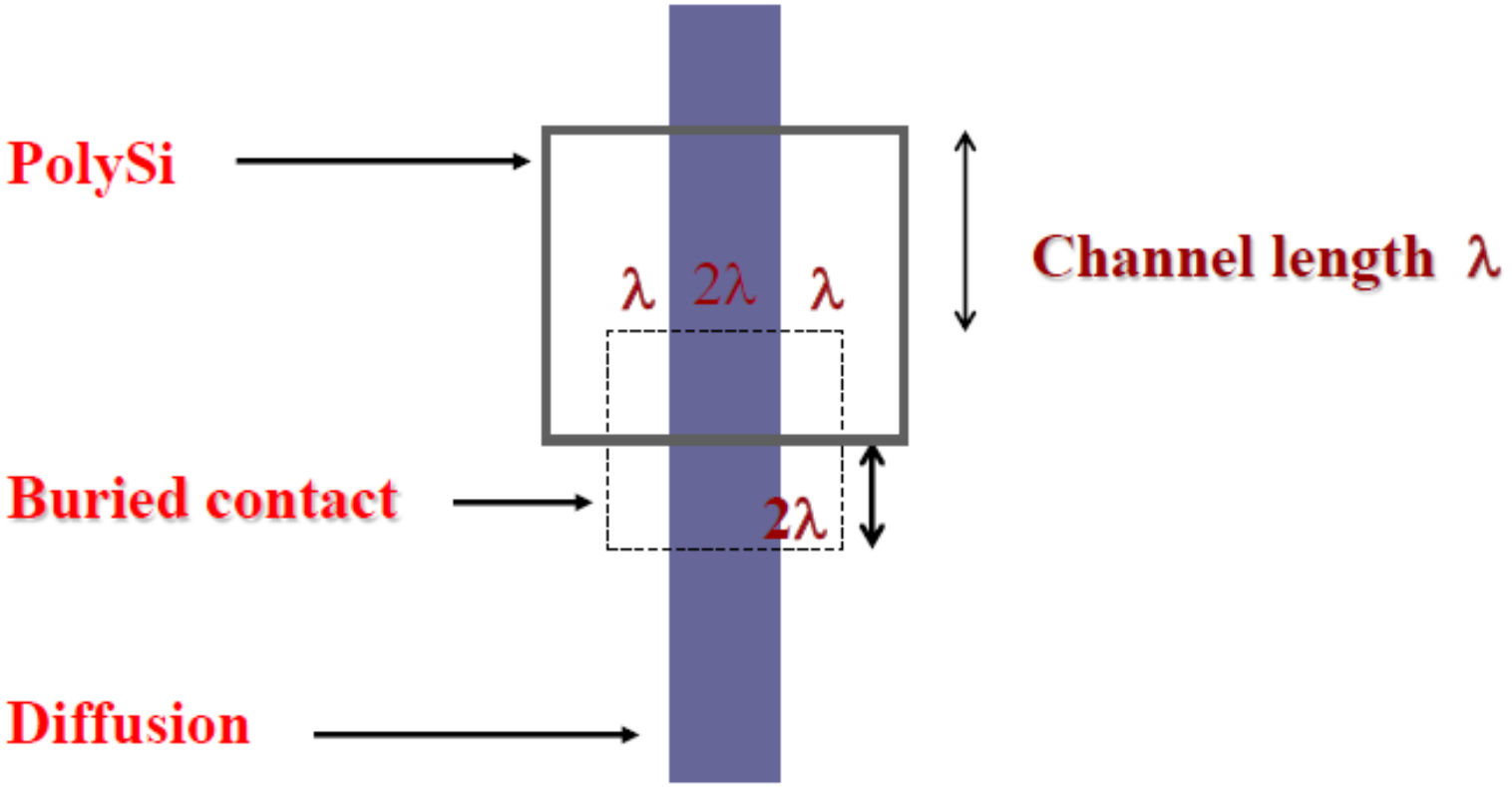
- Diff extend beyond poly by 2λ
- Contact size 2λ * 2λ
- Contact diff/poly/metal overlap 1λ
- Contact to contact spacing 2λ
- Contact to poly/diff spacing 2λ
- Buried contact to active device spacing 2λ
- Buried contact overlap in diff direction 2λ
- Buried contact overlap in poly direction 1λ
- Implant gate overlap 2λ

Buried Contact

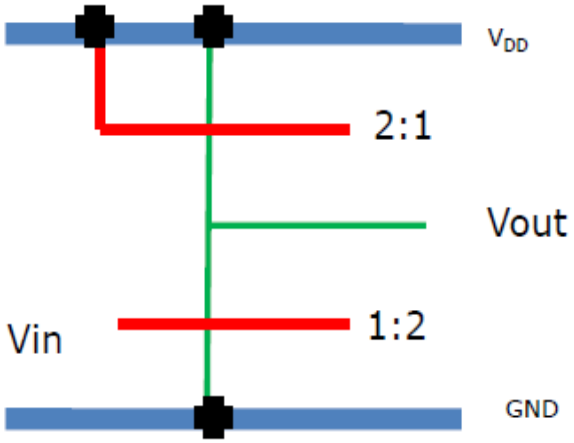
- The buried contact window defines the area where oxide is to be removed so that polySi connects directly to diffusion.
- Contact Area must be a min. of 2 λ *2λ to ensure adequate contact area.



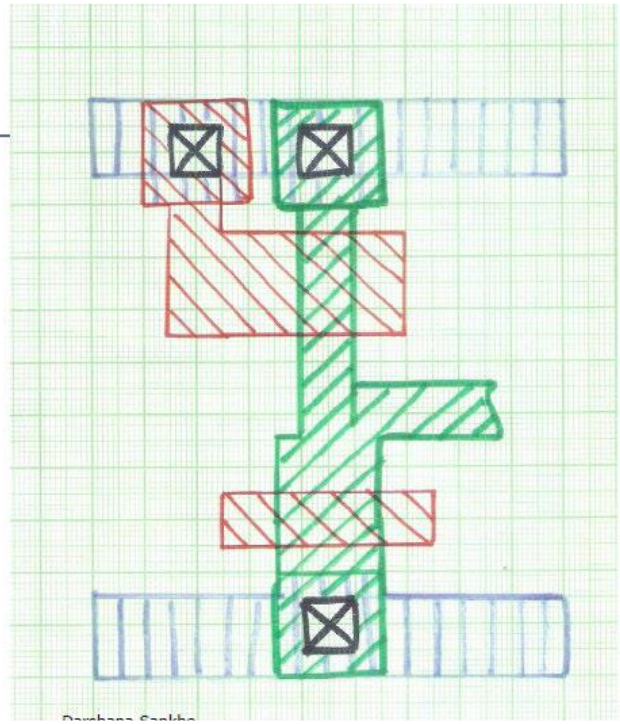
Buried Contact



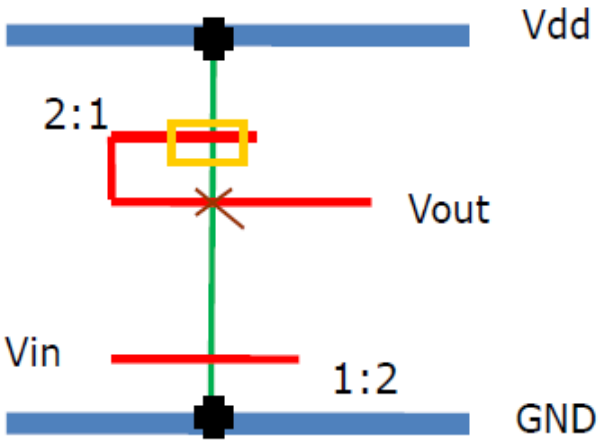
NMOS Inverter: Enhancement load (Stick diagram)



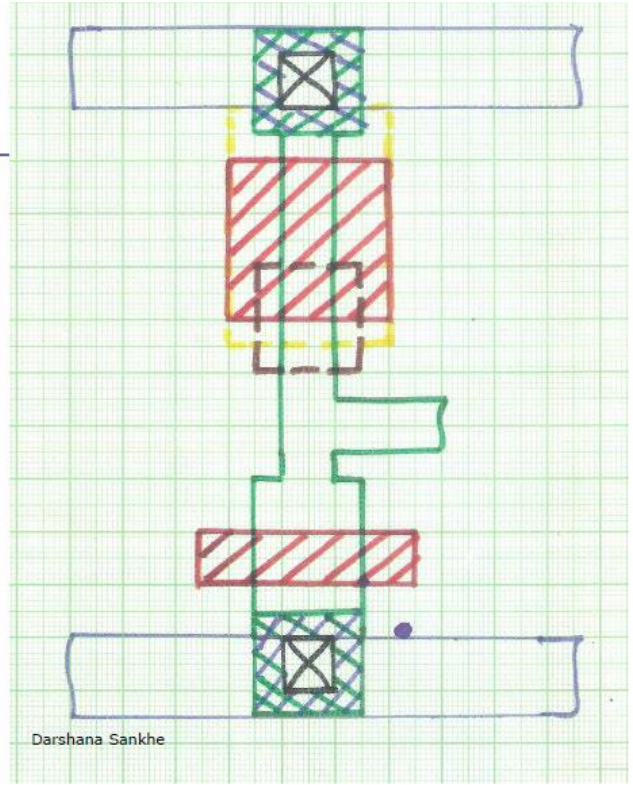
NMOS INVERTER-
Enhancement load



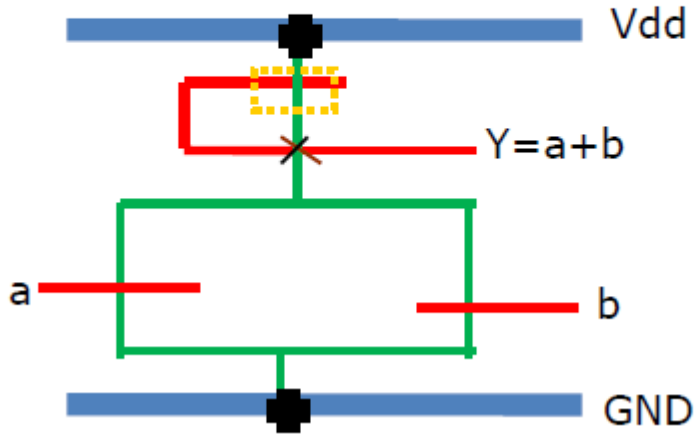
NMOS Inverter: Depletion load (Stick Diagram)



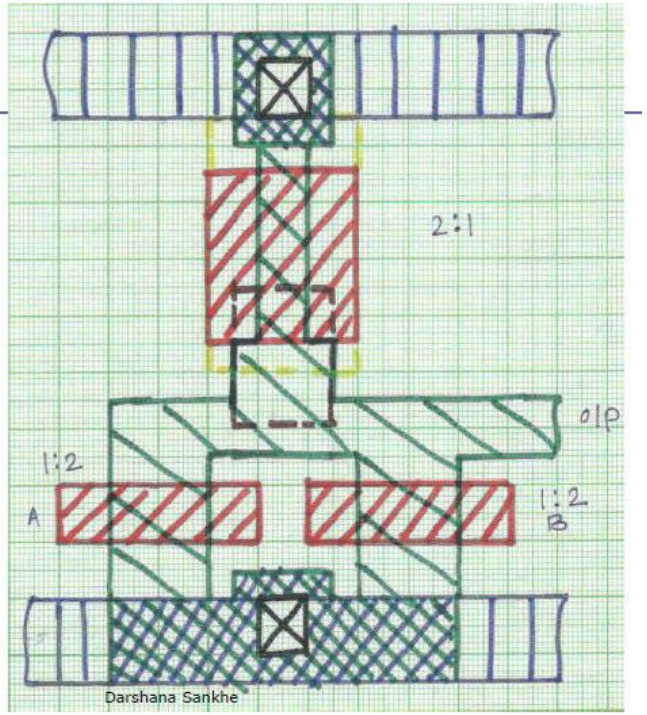
NMOS INVERTER
Depletion load



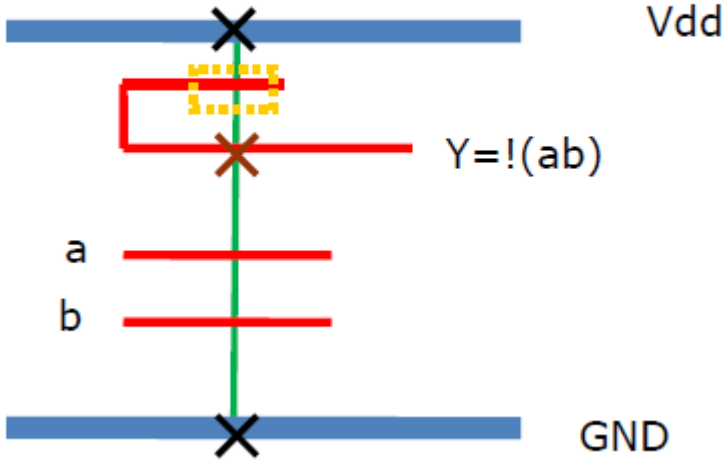
NMOS NOR (Stick Diagram)



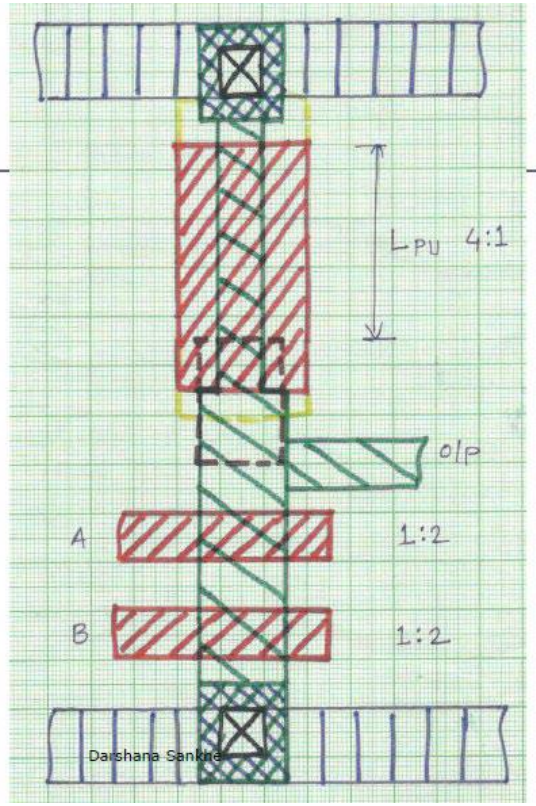
NMOS NOR



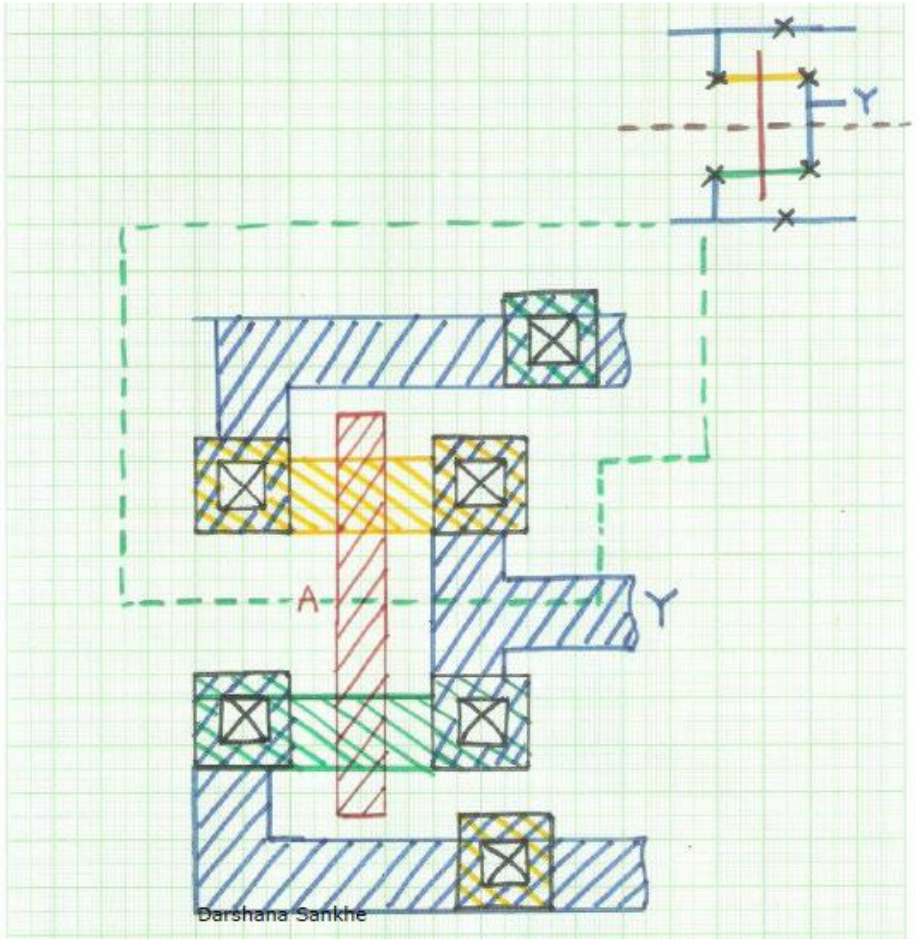
NMOS NAND (Stick Diagram)

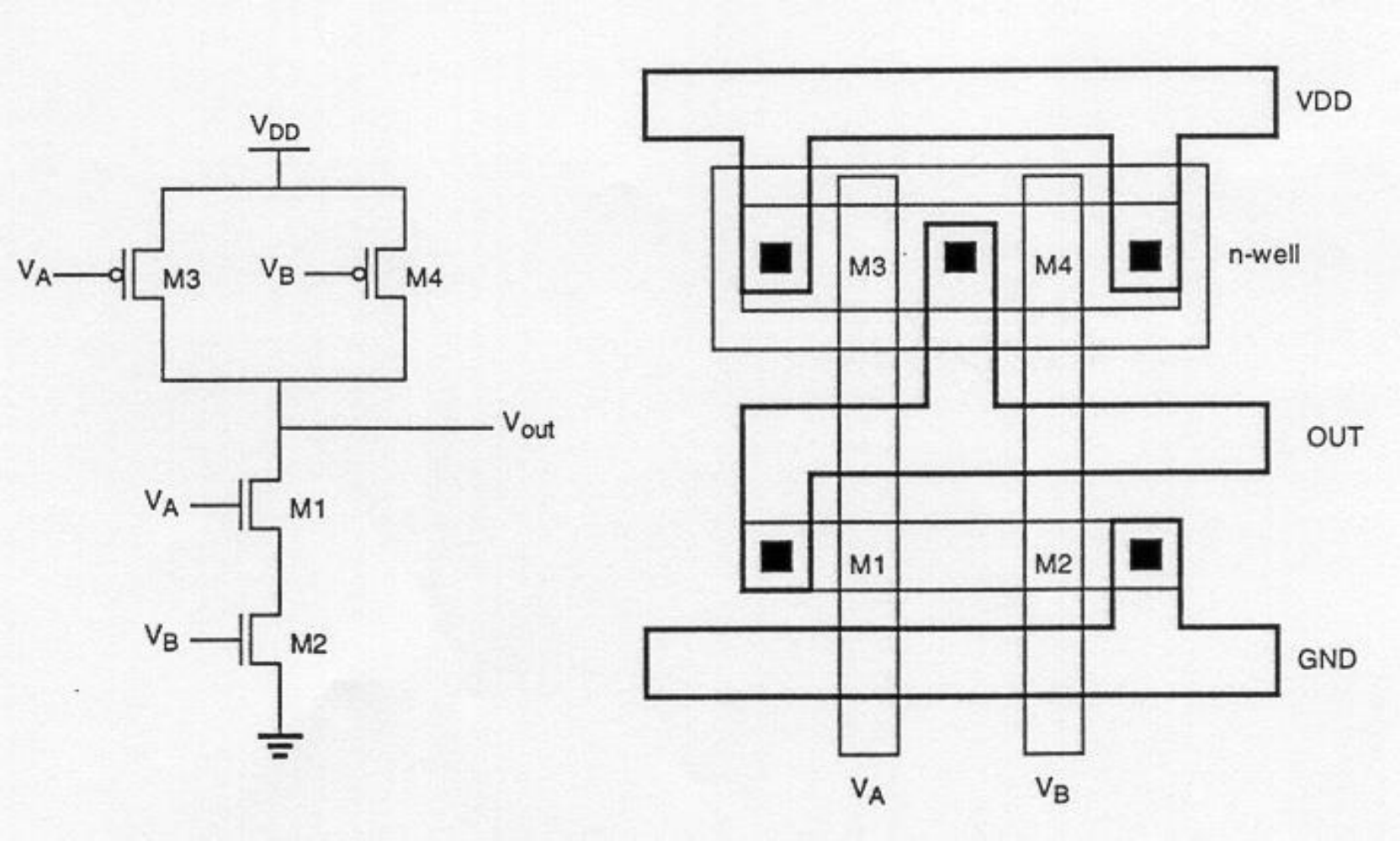


NMOS NAND

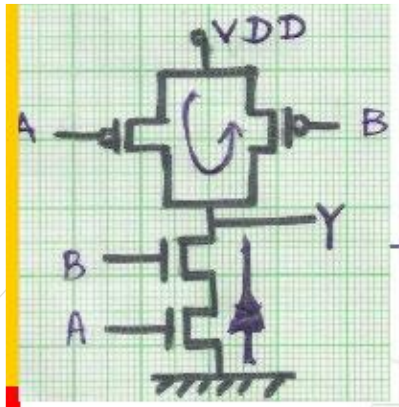


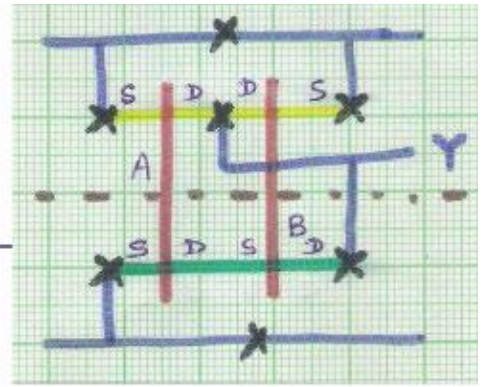
CMOS Inverter:

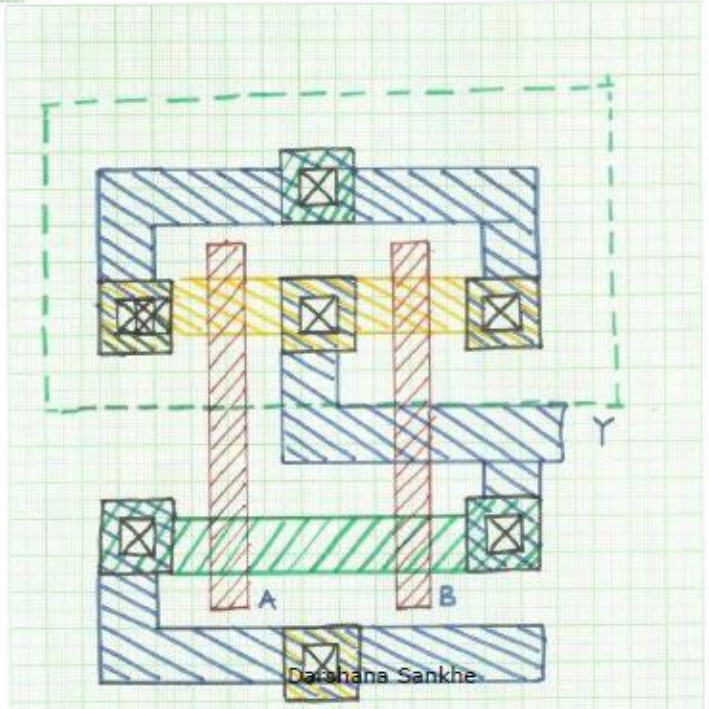


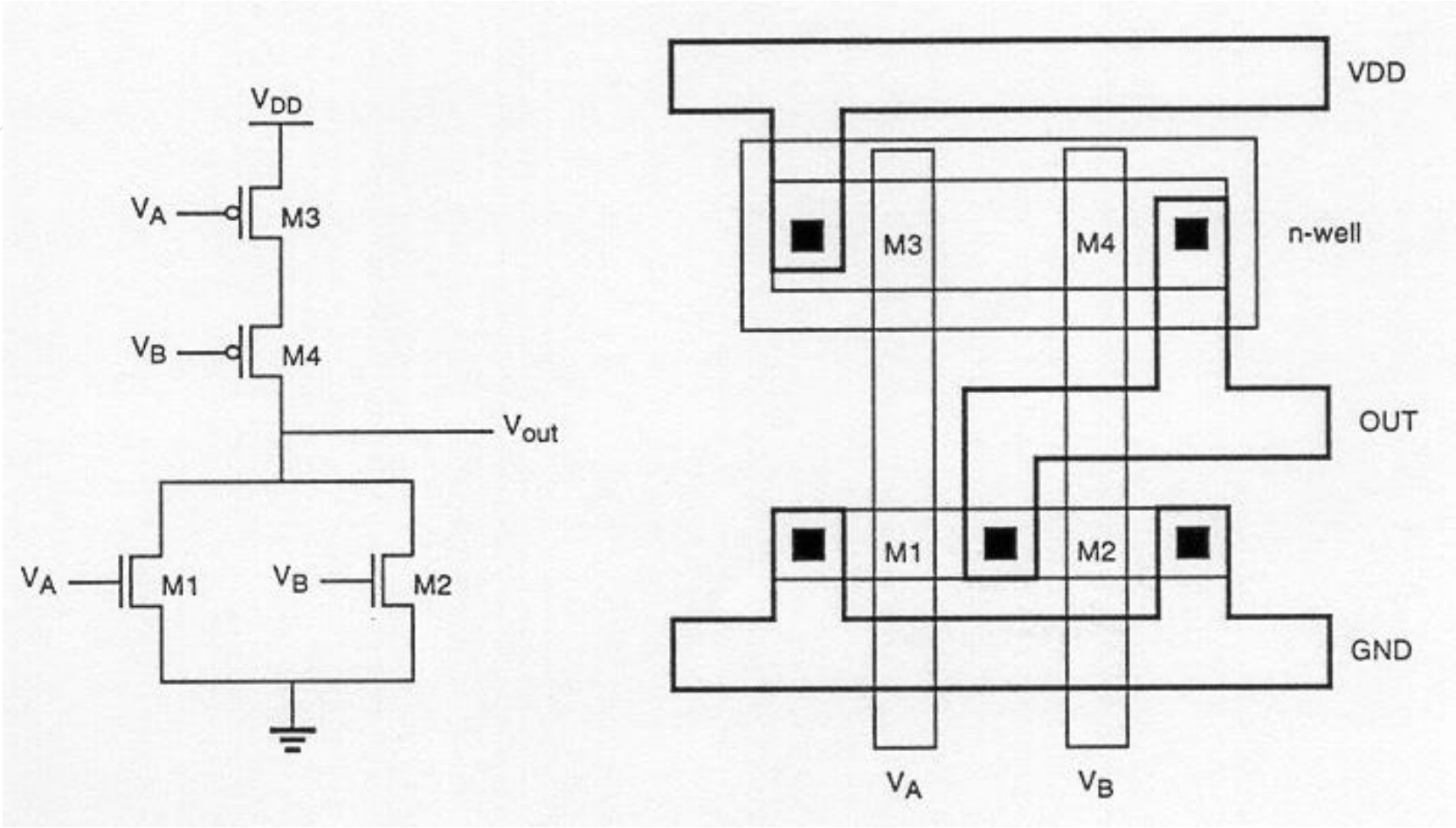

VDD
VA
M3
VB
M4
Vout
VA
M1
VB
M2
VDD
n-well
M3
M4
OUT
M1
M2
GND
VA
VB

CMOS NAND

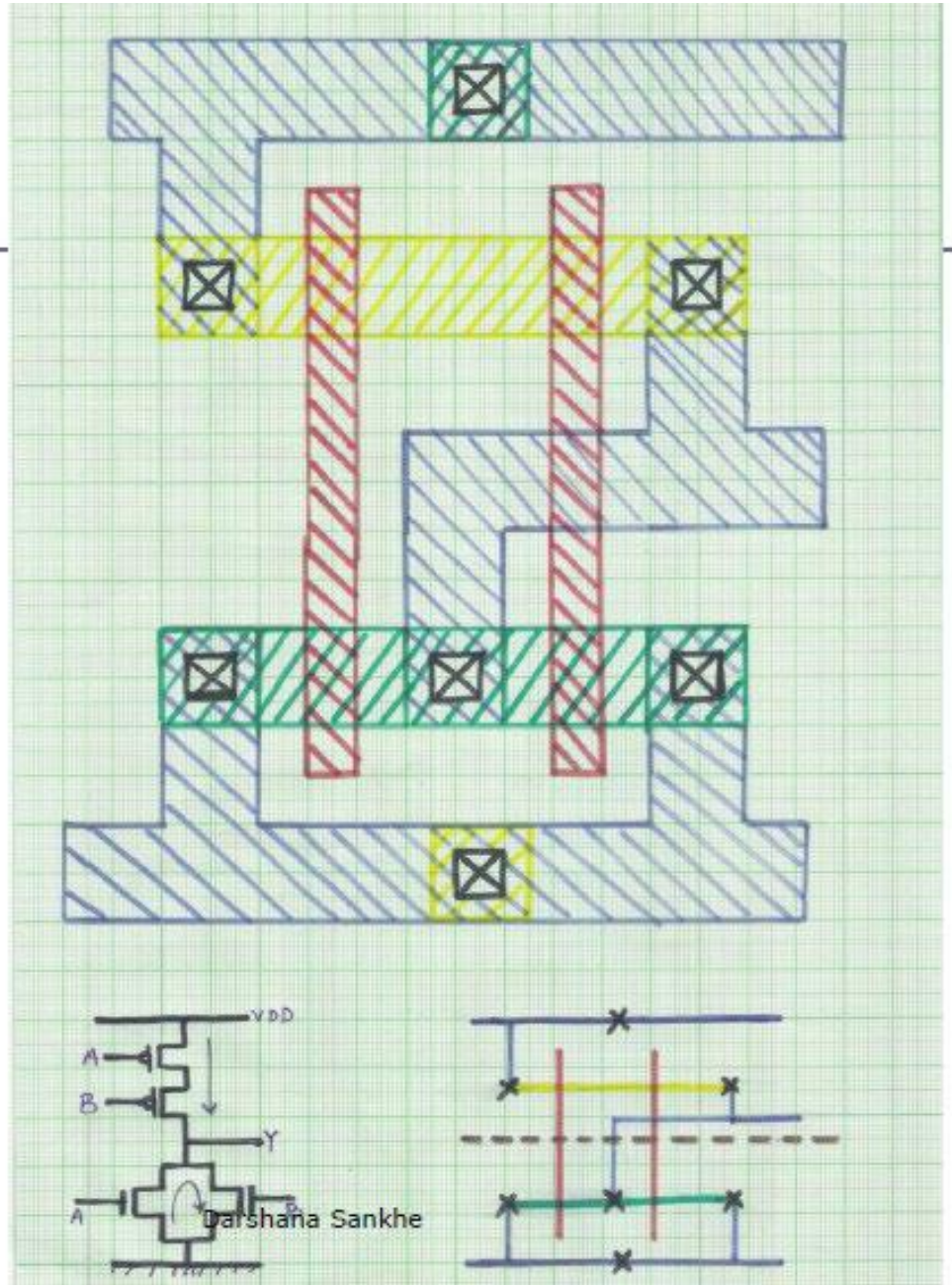


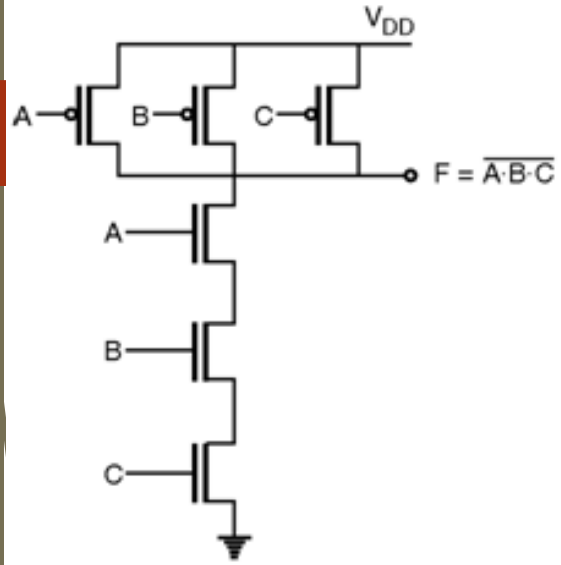





VDD
VA
M3
VB
M4
Vout
VA
M1
VB
M2
VDD
M3
M4
n-well
OUT
M1
M2
GND
VA
VB

CMOS NOR :





(a) Schematic

